

EMT 3102: SOLID AND STRUCTURAL MECHANICS I

B.Sc. Mechatronics Engineering

Continuous Assessment Test II

July 16, 2025 Duration: 1 Hour Total: 30 Marks

Instructions:

- ✧ There are two (2) problems to solve. Each problem carries fifteen (15) marks.
- ✧ You have one (1) hour to complete this test.
- ✧ Show all your working clearly and state any assumptions made.
- ✧ Use appropriate units throughout your calculations.

Problem 1: Torsion of Circular Shafts

[15 marks]

- (a) A solid circular shaft of diameter 115 mm is subjected to a torque of 30 kNm. The angle of twist over a length of 4 m is observed to be 0.1 rad. Determine the modulus of rigidity of the material. $\theta = \frac{TL}{GJ}$ [3 marks]
- (b) A hollow circular shaft has an external diameter of 125 mm and the internal diameter is three-fourths of the external diameter. If the shear stress on a fiber is 37 MPa due to the applied torque, T :
- (i) Calculate the applied torque T [3 marks]
 - (ii) Determine the maximum shear stress [3 marks]
 - (iii) Calculate the angle of twist per unit length if $G = 80$ GPa [3 marks]
- (c) A solid circular shaft has a diameter of 85 mm. Find the maximum shear stress and the twist angle over a length of 2.5 m when the shaft is subjected to a torque of 15 kNm. Take $G = 85$ GPa. [3 marks]

Problem 2: Thin-Walled Pressure Vessels

[15 marks]

- (a) A cylindrical pressure vessel 2.5 m long and 515 mm in diameter with plates 15 mm thick is subjected to an internal pressure of 3.5 MPa. Taking $E = 200$ GPa and Poisson's ratio $\nu = 0.3$ for the vessel material:
- (i) Calculate the circumferential and longitudinal strains [3 marks]
 - (ii) Determine the original volume and change in volume of the vessel [3 marks]
- (b) A cylindrical thin drum 815 mm in diameter and 4.5 m long is made of 15 mm thick plates. The drum is subjected to an internal pressure of 2.5 MPa. Taking $E = 200$ GPa and Poisson's ratio $\nu = 0.25$, calculate the change in diameter and length of the drum. [3 marks]
- (c) A cylindrical shell of diameter 510 mm is required to withstand an internal pressure of 5 MPa. Find the minimum thickness of the shell, if the maximum tensile strength of the plate material is 110 MPa, the efficiency of the joints is 70%, and the factor of safety is 5. [3 marks]
- (d) A cylindrical shell of 1.5 m diameter is made up of 20 mm thick plates. It is subjected to an internal pressure of 2.5 MPa. Taking the efficiency of the joints to be 75%, calculate the circumferential and longitudinal stresses in the shell. [3 marks]

End of Examination
Good Luck!

$\sigma_{avo} = \frac{\sigma_{axial}}{FOS}$